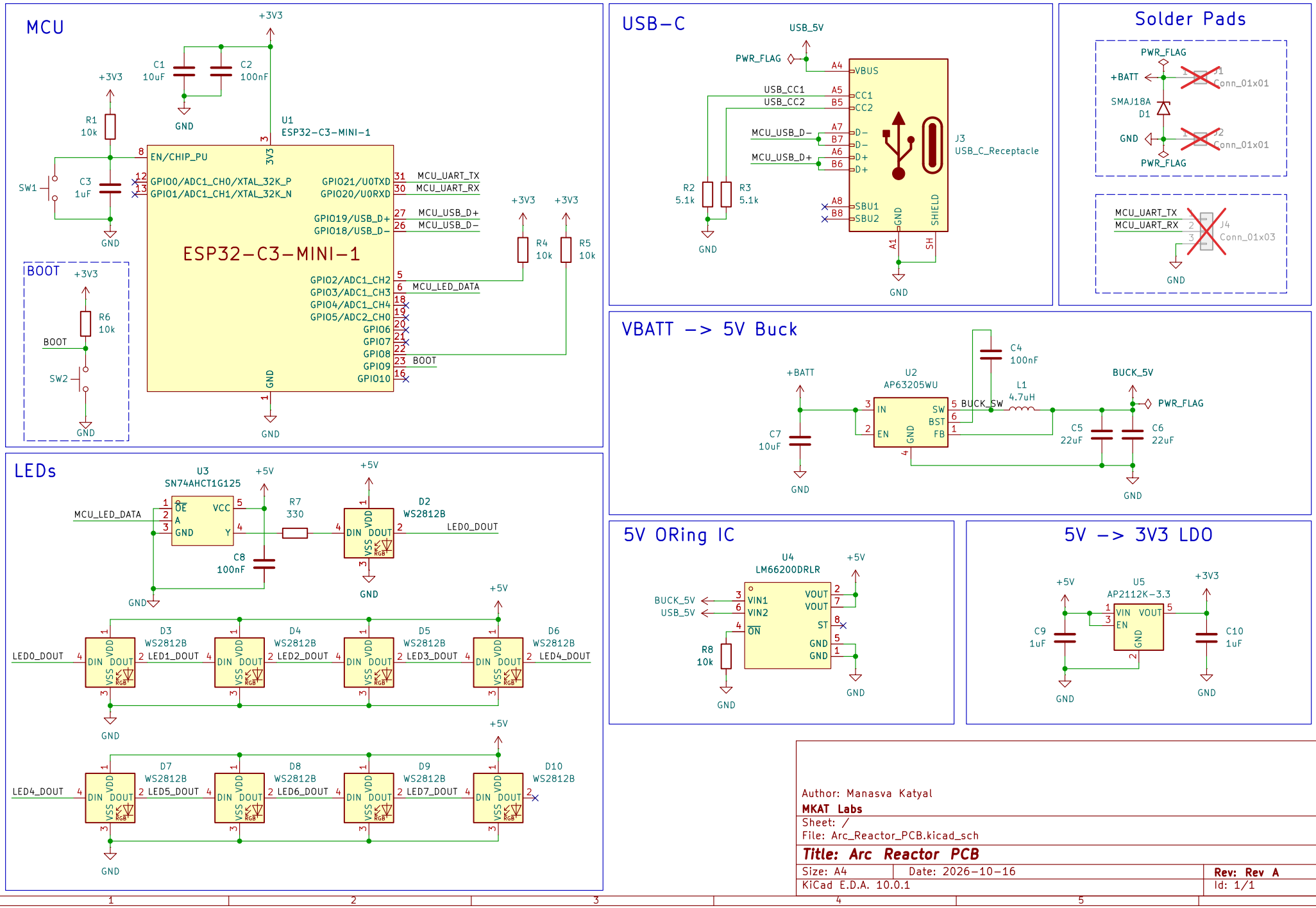

MCU
+3V3
C1
10uF
C2
100nF
GND
R1
10k
U1
ESP32-C3-MINI-1
EN/CHIP_PU
3V3
SW1
C3
1uF
GPIO0/ADC1_CH0/XTAL_32K_P
GPIO1/ADC1_CH1/XTAL_32K_N
GPIO21/U0TXD
GPIO20/U0RXD
MCU_UART_TX
MCU_UART_RX
GPIO19/USB_D+
GPIO18/USB_D-
MCU_USB_D+
MCU_USB_D-
ESP32-C3-MINI-1
R4
10k
R5
10k
GPIO2/ADC1_CH2
GPIO3/ADC1_CH3
GPIO4/ADC1_CH4
GPIO5/ADC2_CH0
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
MCU_LED_DATA
BOOT
BOOT
+3V3
R6
10k
SW2
GND
USB-C
USB_5V
PWR_FLAG
VBUS
USB_CC1
USB_CC2
CC1
CC2
MCU_USB_D-
MCU_USB_D+
D-
D+
J3
USB_C_Receptacle
R2
5.1k
R3
5.1k
SBU1
SBU2
GND
SHIELD
Solder Pads
PWR_FLAG
+BATT
SMAJ18A
D1
J1
Conn_01x01
J2
Conn_01x01
MCU_UART_TX
MCU_UART_RX
J4
Conn_01x03
VBATT -> 5V Buck
+BATT
U2
AP63205WU
IN
EN
SW
BST
FB
BUCK_SW
C4
100nF
L1
4.7uH
BUCK_5V
C5
22uF
C6
22uF
C7
10uF
LEDs
U3
SN74AHCT1G125
MCU_LED_DATA
OE
A
VCC
Y
R7
330
C8
100nF
+5V
D2
WS2812B
LED0_DOUT
DIN
DOUT
VDD
VSS
D3
WS2812B
LED1_DOUT
D4
WS2812B
LED2_DOUT
D5
WS2812B
LED3_DOUT
D6
WS2812B
LED4_DOUT
D7
WS2812B
LED5_DOUT
D8
WS2812B
LED6_DOUT
D9
WS2812B
LED7_DOUT
D10
WS2812B
5V ORing IC
U4
LM66200DRLR
BUCK_5V
USB_5V
VIN1
VIN2
ON
VOUT
ST
GND
R8
10k
5V -> 3V3 LDO
U5
AP2112K-3.3
VIN
VOUT
EN
C9
1uF
C10
1uF
Author: Manasva Katyal
MKAT Labs
Sheet: /
File: Arc_Reactor_PCB.kicad_sch
Title: Arc Reactor PCB
Size: A4
Date: 2026-10-16
Rev: Rev A
KiCad E.D.A. 10.0.1
Id: 1/1